

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	ITA0512	Date:	21/08/2026

Code of Conduct

I Yavanika S [192521258] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: YAVANIKA S

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Question 1: Detection Performance Comparison

Two object detection models produce the following results — compute precision and recall for both models and compare their performance.

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Precision and Recall Calculation

$Precision = TP / (TP + FP)$ $Recall = TP / (TP + FN)$

Model A — Viola-Jones

$Precision = 80 / (80 + 20) = 80 / 100 = 0.80 \rightarrow 80.00\%$
 $Recall = 80 / (80 + 30) = 80 / 110 = 0.7273 \rightarrow 72.73\%$
 $F1-Score = 2 \times (0.80 \times 0.7273) / (0.80 + 0.7273) = 0.7619 \rightarrow 76.19\%$

Model B — YOLO

$Precision = 90 / (90 + 15) = 90 / 105 = 0.8571 \rightarrow 85.71\%$
 $Recall = 90 / (90 + 20) = 90 / 110 = 0.8182 \rightarrow 81.82\%$
 $F1-Score = 2 \times (0.8571 \times 0.8182) / (0.8571 + 0.8182) = 0.8372 \rightarrow 83.72\%$

Model	Precision	Recall	F1-Score
A (Viola-Jones)	80.00%	72.73%	76.19%
B (YOLO)	85.71%	81.82%	83.72%

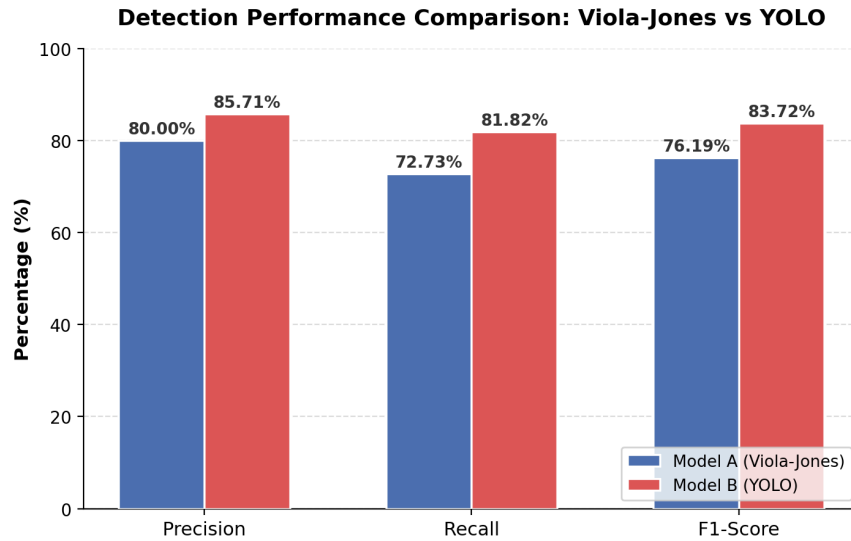


Fig 1.1 — Precision, Recall and F1-Score comparison of Model A vs Model B

Flowchart: Detection Performance Evaluation

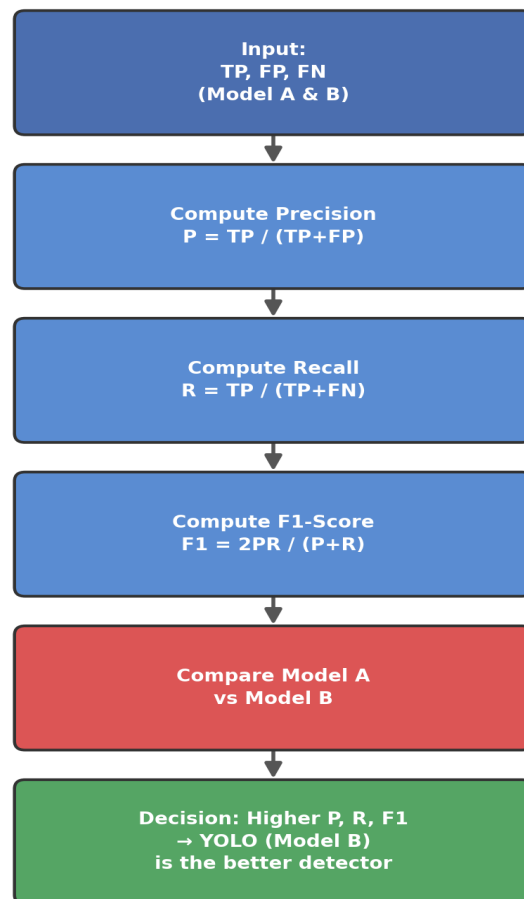


Fig 1.2 — Evaluation pipeline flowchart for object detection comparison

(b) Comparison and Conclusion

- Precision: Model B (85.71%) is higher than Model A (80.00%) — YOLO produces fewer false alarms (15 FP vs 20 FP).
- Recall: Model B (81.82%) is higher than Model A (72.73%) — YOLO misses fewer actual objects (20 FN vs 30 FN).
- F1-Score: Model B (83.72%) outperforms Model A (76.19%), confirming a better balance between precision and recall.
- Viola-Jones uses hand-crafted Haar features and a cascade classifier, which struggles with scale, pose, and lighting variation.
- YOLO uses a deep CNN with learned features and single-shot detection, giving stronger generalization on real-world data.

Final Answer: Model B (YOLO) is the better object detection model — it scores higher on Precision, Recall and F1-Score than Model A (Viola-Jones).

Question 2: Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors (in pixels) — compute the average error for both methods and evaluate which is more accurate.

Frame	Method A (px)	Method B (px)
1	2	3
2	3	5
3	2	4

(a) Average Error Calculation

Average Error = (Sum of frame errors) / (Number of frames)

Method A

Average Error (A) = (2 + 3 + 2) / 3 = 7 / 3 = 2.33 pixels

Method B

Average Error (B) = (3 + 5 + 4) / 3 = 12 / 3 = 4.00 pixels

Method	Average Displacement Error
A	2.33 pixels
B	4.00 pixels

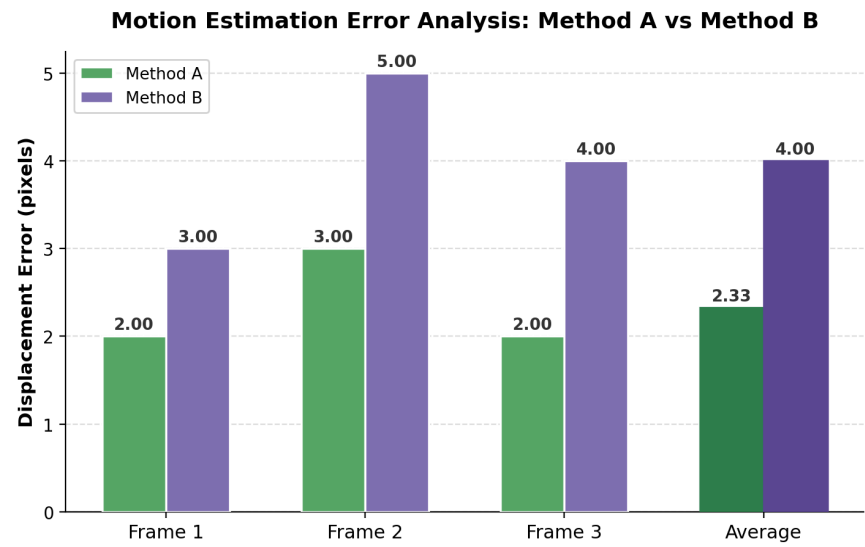


Fig 2.1 — Frame-wise displacement error and average error: Method A vs Method B

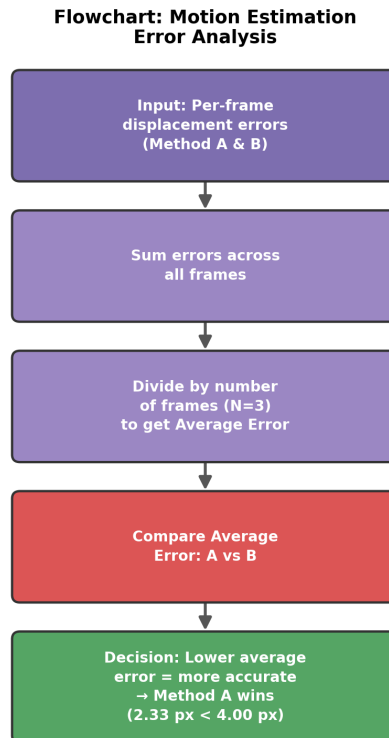


Fig 2.2 — Motion estimation error-analysis flowchart

(b) Comparison and Evaluation

- Method A shows consistently smaller displacement errors across all 3 frames (2, 3, 2 px).
- Method B shows consistently larger displacement errors across all 3 frames (3, 5, 4 px).
- The average error of Method A (2.33 px) is 1.67 px lower than Method B (4.00 px) — around 42% lower.
- A lower displacement error means the estimated motion vectors are closer to the true (ground-truth) motion, i.e. higher tracking accuracy.

Final Answer: Method A is the more accurate motion estimation technique — it produces a lower average displacement error (2.33 px) than Method B (4.00 px) across every frame.

Question 3: Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame) are measured across three image regions — compute the average motion magnitude for both methods and evaluate which captures stronger motion.

Region	Method A (px/frame)	Method B (px/frame)
R1	5	8
R2	6	10
R3	5	9

(a) Average Motion Magnitude Calculation

Average Magnitude = (Sum of region magnitudes) / (Number of regions)

Method A

Average Magnitude (A) = (5 + 6 + 5) / 3 = 16 / 3 = 5.33 pixels/frame

Method B

Average Magnitude (B) = (8 + 10 + 9) / 3 = 27 / 3 = 9.00 pixels/frame

Method	Average Optical Flow Magnitude
A	5.33 px/frame
B	9.00 px/frame

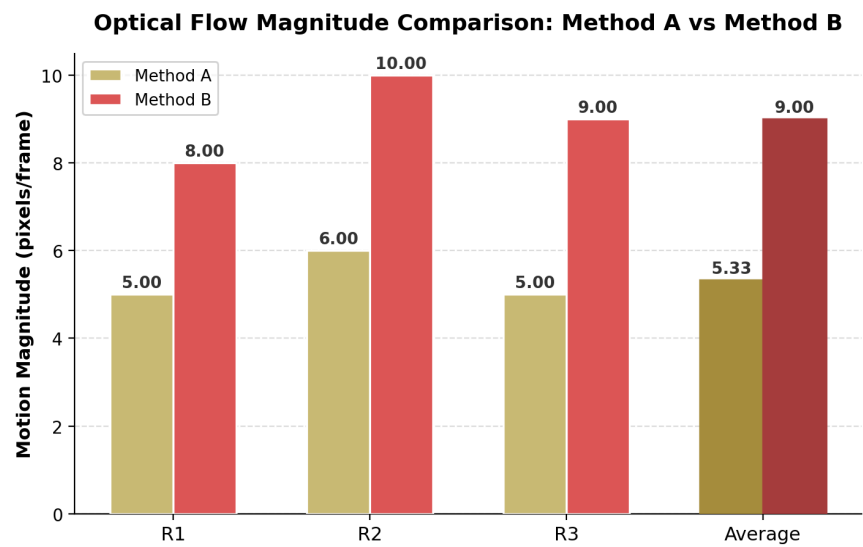


Fig 3.1 — Region-wise optical flow magnitude and average: Method A vs Method B

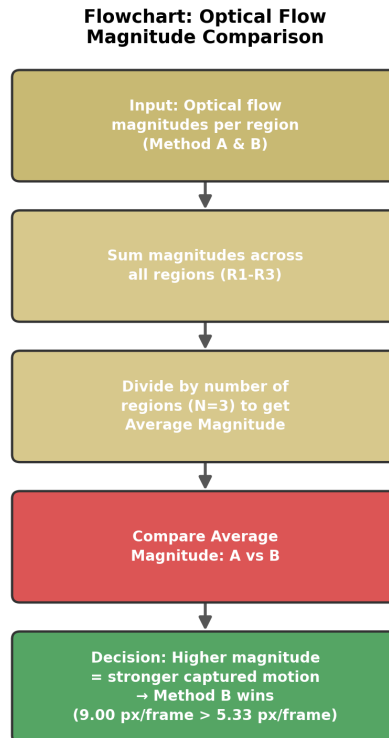


Fig 3.2 — Optical flow magnitude comparison flowchart

(b) Comparison and Evaluation

- Method B records a higher magnitude than Method A in every region (R1: 8 vs 5, R2: 10 vs 6, R3: 9 vs 5).
- The average magnitude of Method B (9.00 px/frame) is about 69% higher than Method A (5.33 px/frame).
- Higher optical flow magnitude means the technique is registering larger pixel displacement, i.e. it is more sensitive to motion.
- A higher magnitude is only 'better' if it reflects true scene motion rather than noise — Method B should be checked against ground truth to rule out over-estimation.

Final Answer: Method B captures stronger (larger-magnitude) motion than Method A in every region, with a higher overall average (9.00 vs 5.33 px/frame).